

A New Distributed Optimal Utility Max-Min Fair Resource Allocation

Jian Zhao, Danni Song, Qiang Zhang, Jintao Meng

Abstract—A central issue in computer network is how to allocate bandwidth to flows fairly, in a decentralized manner. However, there are few researches in the literature of utility-fair network resource allocation algorithm. In this paper, we propose a new approach to achieve utility max-min fairness in network bandwidth allocation, which based on a vector optimization and some extension to the famous kuhn-tucker conditions for nonlinear optimization. We formulate a nonlinear program that finds the optimal bandwidth allocation. This leads to a source link algorithm which we show its convergence to the optimal allocation by an example.

I. INTRODUCTION

Max-min fairness is a simple, widely used approach to define fairness in networks [1]. It results in higher average throughput and better utilization of the resources than *equal sharing* policy of the resources. There are many works that provide centralized or distributed max-min flow control algorithms that work in single or multiple bottleneck networks in spite of round-trip delays, such as [2][3][4]. But this fairness criterion just takes into account the bandwidth allocated to each application, with no regard to differences in applications. It assumes that the benefit to each application is the same for a given throughput. Actually, in real networks, different applications have their own specific demands. For example, real-time applications require strict bandwidth, delay and delay jitter guarantee. On the other hand, data applications do not require strict delay jitter guarantee. To take application demands into consideration, Shenker [5] introduced the utility function f_i , which maps the service s_i delivered by the network into the performance of application i . In general, the quantity s_i contains all measures of service delivered to an application, such as, bandwidth, delay, delay jitter and some other factors. In this paper, we will focus on the utility as a function of the allocated bandwidth.

Following the concept of utility function, there have been many papers providing utility-based network bandwidth allocation schemes based on optimization theory [6][7][8]. In these papers, they all maximize the aggregation of users' utilities under capacity constraint, but they differ in the

solution approaches which lead to different flow control algorithms, which in turn lead to different marking implementations of the algorithms. The approach in [6] decomposes the original system problem into user problem and network problem, by solving these two sub-problems it is easy to construct a distributed primal-dual algorithm to solve the system problem. The approach in [7] solves the dual of the original optimization problem using gradient projection algorithm. In [8], each user controls its window size in stead of its rate, based on the total delay, so it can be viewed as a refinement of TCP congestion control algorithm. Although these frameworks achieved the maximization of aggregate utility, there is relatively less utility-aware fairness consideration. The bandwidth proportional fairness for logarithmic utility functions and arbitrarily close approximation to bandwidth max-min fairness for a generalized utility function which is concave are respectively achieved in [6] and [8].

On the other hand, Cao and Zegura [9] defined a utility max-min fairness and proposed a bandwidth allocation scheme which achieves this fairness. Although their work is the first step to the utility max-min flow control, it requires that each link knows the utility functions of all the flows sharing the link, which is not realistic.

To overcome this limitations, Hyang-Won Lee et al.[10] proposed a utility max-min flow control algorithm based on optimization theory. Their algorithm operates in an asynchronous and distributed manner and does not require any per-flow operation in the network. They mathematically proved that their proposed algorithm converges to a unique global optimal point under some conditions in the case of a single bottleneck link. Through simulation, they also deduced the algorithm works well for the case of multiple bottleneck links. But in their problem formulation and distributed algorithm presentation they implicitly assumed that each link knows the utility functions of all the flows sharing the link. Obviously, this is not realistic, since each user has nothing about the utility function of other users'. So in this strict sense this algorithm is not distributed because they require global information, such as utility functions of users'.

Our work is based on optimizing the utility vector other than aggregation of utilities at a given rate allocation. By transforming the original problem in to some equivalent conditions and using gradient projection algorithm, we can get the algorithm of the sources and links.

The rest of the paper is organized as follows. In Section 2, we formulate the utility max-min flow control problem. And then, some theorems to solve this problem is given and proved in Section 3. In Section 4, we propose the source link algorithm. We give an example in Section 5 and conclude the paper in Section 6.

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II. PROBLEM FORMULATION

Consider a network consisting of a set L of links of capacities $c_l, l \in L$. It is shared by a set S of sources. Each source s attains a strictly increasing and continuously differentiable utility function of $U_s(x_s)$ when it transmits at rate x_s which satisfies $0 \leq m_s \leq x_s \leq M_s < \infty$, where m_s and M_s are minimum and maximum rates required by source s . Let $L_s \subseteq L$ be a subset of links that connect sources s to the destination. For convenience, let $S_l = \{s \in S \mid l \in L_s\}$ denote the set of sources that use link l . Note that $l \in L_s$ if and only if $s \in S_l$. Define the matrix $A = (A_{ji}, j \in L, i \in S)$ where $A_{ji} = 1$ if connection i uses link j and $A_{ji} = 0$, otherwise. This defines an adjacency matrix A . Let also $A_{\bullet i} = \{j \mid A_{ji} = 1\}$ be the set of links that connection i uses, $A_{j\bullet} = \{i \mid A_{ji} = 1\}$ the set of connections that use link j and $C = (c_l, l \in L)$.

DEFINITION 1 (Utility max-min fairness) A feasible allocation of rate x is utility max-min fair if and only if an increase of any rate x_i with utility $U_i(x_i)$ within the domain of feasible allocations must be at the cost of a decrease of some rate x_j which satisfies $U_j(x_j) \leq U_i(x_i)$. Formally, for any other feasible allocation y , if $U_s(y_s) > U_s(x_s)$ for some s , then there must exist some t such that $U_t(y_t) < U_t(x_t) \leq U_s(x_s)$.

The definition above is similar to the definition of bandwidth max-min fairness except that in this definition, utilities are compared instead of bandwidths.

Similar to the bandwidth max-min allocation, the utility max-min fair allocation can be computed by using "progressive filling". The idea is as follows. Start with all rates equal to 0 and all source utilities equal to 0, then grow all rates together at the same utility pace u , that is, for each link $l \in L$, $\forall i, j \in S_l$ $U_i(u) = U_j(u)$

$\sum_{i \in S_l} x_i \leq c_l$, when at some time the capacity limit is

reached at some link, freeze the rate allocation of those sources that go through this link, but proceed with this filling for other sources. All the sources that are stopped have a bottleneck link. This is because they use a saturated link, and all other sources using the saturated link are stopped at the same time, or were stopped before, thus have a smaller or equal utility. The algorithm continues until it is not possible to increase. The algorithm terminates because L and S are finite.

Our objective is to choose source rates $x = (x_s, s \in S)$ so as to solve problem

$$\begin{aligned} \text{P: } \max \min [U_1(x_1), U_2(x_2), \dots, U_s(x_s)] \\ \text{subject to } Ax \leq C \end{aligned}$$

Though this problem cannot be mathematically solved, it can be transformed equivalently to some conditions. Through these conditions we can get some protocols for problem P, which we use in bandwidth allocation to adjust source rates. In section 3, we give some related theorems and details to transform the problem.

III. THEOREMS AND PROOFS

Let $x \rightarrow u = F(x)$ be a differentiable mapping of nonnegative n-vectors x into m-vectors u . That is, $F(x)$ is an m-vector whose components $f_1(x), \dots, f_m(x)$ are differentiable functions of x defined for $x \geq 0$. Let $x \rightarrow v = U(x)$ be a differentiable mapping of nonnegative n-vectors x into p-vectors v . That is, $U(x)$ is a p-vector whose components $U_1(x), \dots, U_p(x)$ are differentiable functions of x defined for $x \geq 0$. Taking partial derivatives, evaluated at x^0 , let

$$F^0 = [\partial f_h / \partial x_i]^0 \quad U^0 = [\partial U_h / \partial x_i]^0.$$

Here F^0 is an m by n matrix and U^0 is a p by n matrix. Let u_k^0 denote the n-vector whose components form the k -th row of U^0 .

Let $\min[U(x)]$ denote the function whose value for each $x \geq 0$ is the least among the p values $U_1(x), \dots, U_p(x)$ of the components of the vector $U(x)$.

Minimum component maximum problem. To find an x^0 that maximizes

$$\min[U(x)] \text{ constrained by } F(x) \geq 0, x \geq 0.$$

To solve this problem we need the following three theorems. Before giving these theorems and their proofs, we first present three conditions and constraint qualification.

Let $\phi(x, w)$ be a differentiable function of an n-vector x with components $x_i \geq 0$ and an m-vector w with components $w_h \geq 0$. Taking partial derivatives, evaluated at a particular point x^0, w^0 , let

$$\phi_x^0 = [\partial \phi / \partial x_i]^0 \quad \phi_w^0 = [\partial \phi / \partial w_h]^0.$$

Here ϕ_x^0 is an n-vector and ϕ_w^0 is an m-vector.

$$\text{Condition (1)} \quad \phi_x^0 \leq 0, \quad \phi_x^0 x^0 = 0, \quad x^0 \geq 0$$

$$\text{Condition (2)} \quad \phi_w^0 \geq 0, \quad \phi_w^0 w^0 = 0, \quad w^0 \geq 0$$

$$\text{Condition (3)} \quad \phi(x, w^0) \leq \phi(x^0, w^0) + \phi_x^0 (x - x^0)$$

Constraint qualification. Let x^0 belong to the boundary of the constraint set of points x satisfying $F(x) \geq 0, x \geq 0$. Let the inequalities $F(x^0) \geq 0, Ix^0 \geq 0$ (where I is the identity matrix of order n) be separated in to $F_1(x^0) = 0, I_1x^0 = 0$ and $F_2(x^0) > 0, I_2x^0 > 0$. It will be assumed for each x^0 of the boundary of the constraint set that any vector differential dx satisfying the homogeneous linear inequalities $F_1^0 dx \geq 0, I_1 dx \geq 0$ is tangent to an arc contained in the constraint set; that is, to any dx satisfying $F_1^0 dx \geq 0, I_1 dx \geq 0$ there corresponds a differentiable arc $x = a(\theta), 0 \leq \theta \leq 1$, contained in the constraint set, with $x^0 = a(0)$, and some positive scalar λ such that $[da/d\theta]^0 = \lambda dx$. This qualification is designed to rule out singularities on the boundary of the constraint set. It is required in the following theorems' proof. But it is not required for our algorithm, because in our utility max-min problem, the feasible region is a convex set formed by linear constraints only, then the constraint qualification will invariably be met, and the theorems will always hold at an optimal solution.

Theorem 1(necessary condition)

In order that x^0 be a solution of the problem, it is necessary that there be some nonnegative v^0 with unit component sum satisfying $v^0 U(x^0) = \min[U(x^0)]$ and such that x^0 and some w^0 satisfy conditions (1) and (2) for $\phi(x, w) = v^0 U(x) + w' F(x)$.

Proof. Let x^0 be a solution of the problem, then the inequalities $F(x^0) \geq 0, Ix^0 \geq 0$ (where I is the identity matrix of order r) hold. Let the inequalities $F(x^0) \geq 0, Ix^0 \geq 0$ be separated into $F_1(x^0) = 0, I_1x^0 = 0$ and $F_2(x^0) > 0, I_2x^0 > 0$.

Let $U(x^0)$ be separated into $U_1(x^0) = \min[U(x^0)]$ and $U_2(x^0) > \min[U(x^0)]$. Then since x^0 is assumed to maximize $\min[U(x)]$ constrained by $F(x) \geq 0, x \geq 0$, one must have that $U_1^0 dx > 0$ for no vector differential dx satisfying $F_1^0 dx \geq 0, I_1 dx \geq 0$ (or for no dx at all, if F_1^0 and I_1 are vacuous). That is, for each k belonging to a certain non vacuous subset of the set of indices corresponding to the rows of U^0 that belong to U_1^0 one must have that $u_k^0 dx \leq 0$ for all dx satisfying $F_1^0 dx \geq 0, I_1 dx \geq 0, U_1^0 dx \geq 0$. Hence, by the fundamental property of homogeneous linear inequalities that an inequality $b'x \geq 0$ holds for all n -vectors x satisfying a

system of m inequalities $Ax \geq 0$ only if $b = A't$ for some m -vector $t \geq 0$ [12]. Hence

$$-u_k^0 = F_1^0 w_1^k + I_1^0 s_1^k + U_1^0 v^k \text{ for some } w_1^k \geq 0, s_1^k \geq 0, v_1^k \geq 0$$

Now, summing for k over the nonempty subset and transferring the U_1^0 terms to the left side, one has

$$-U_1^0 v_1^0 = F_1^0 w_1^0 + I_1^0 s_1^0, \text{ where } w_1^0 = \sum w_1^k \geq 0, s_1^0 = \sum s_1^k \geq 0, \text{ and } v_1^0 = e_1 + \sum v_1^k \geq 0, e_1 \text{ being a vector whose components are 0's or 1's, with at least one 1. Here it can be assumed that the sum of the components of } v_1^0 \text{ is one, since the above vector equation is homogeneous and the sum of the components of } v_1^0 \text{ is positive. Form } v^0 \text{ from } v_1^0 \text{ by adding zeros as components.}$$

Then $v^0 U(x^0) = v_1^0 U_1(x^0) = \min[U(x^0)]$. By setting

$g(x) = v^0 U(x)$, the above vector equation can be rewritten as

$$-g^0 = -U^0 v_1^0 = -U_1^0 v_1^0 = F_1^0 w_1^0 + I_1^0 s_1^0.$$

By adding zeros as components to w_1^0 and s_1^0 to form w^0 and s^0 , the above equation may be rewritten as $-g^0 = F^0 w^0 + s^0$ for some $w^0 \geq 0, s^0 \geq 0$.

Consequently,

$$\phi_x^0 = g^0 + F^0 w^0 \leq g^0 + F^0 w^0 + s^0 = 0$$

At the same time, since $s^0 x^0 = s_1^0 I_1 x^0 = 0$,

$$\phi_x^0 x^0 = g^0 x^0 + w^0 F^0 x^0 = 0$$

Moreover, $\phi_w^0 = F(x^0) \geq 0$ and

$$\phi_w^0 w^0 = w^0 F(x^0) = w_1^0 F_1(x^0) = 0$$

This completes the proof of the theorem.

Theorem 2 (sufficient condition)

In order that x^0 be a solution of the minimum component maximum problem, it is sufficient that there be some nonnegative v^0 with unit component sum satisfying condition $v^0 U^0(x) = \min[U^0(x)]$ and such that x^0 and some w^0 satisfy conditions (1), (2) and (3) for

$$\phi(x, w) = v^0 U(x) + w' F(x)$$

Proof. From (3), (1), (2) one has that

$$v^0 U(x) + w^0 F(x) = \phi(x, w^0) \leq \phi(x^0, w^0) + \phi_x^0 (x - x^0) \phi(x^0, w^0) = v^0 U(x^0) + w^0 F(x^0) = v^0 U(x^0)$$

But $w^0 F(x) \geq 0$ for all x satisfying $F(x) \geq 0$. Hence

$$v^0 U(x) \leq v^0 U(x^0) \text{ for all } x \text{ satisfying the}$$

constraints $F(x) \geq 0$, $x \geq 0$. But v^0 is nonnegative with components sum and satisfies $v^0 U(x^0) = \min[U(x^0)]$.

$$\min[U(x)] \leq v^0 U(x) \leq v^0 U(x^0) = \min[U(x^0)]$$

for all x satisfying the constraints.

This proves theorem 2.

Theorem 3(Equivalence theorem)

Let the functions $f_1(x), \dots, f_m(x)$ and

$U_1(x), \dots, U_p(x)$ be concave as well as differentiable

for $x \geq 0$. Then, x^0 is a solution of the minimum component maximum problem if and only if, there is some nonnegative v^0 with unit component sum satisfying

condition $v^0 U^0(x) = \min[U^0(x)]$ and such that x^0 and some w^0 satisfying conditions (1), (2) and (3) for

$$\phi(x, w) = v^0 U(x) + w' F(x).$$

Proof. Clearly $g(x) = v^0 U(x)$ is concave, since v^0 is nonnegative. Because of the concavity of $F(x)$ and $U(x)$ [13], we have $F(x) \leq F(x^0) + F'^0(x - x^0)$

and $g(x) \leq g(x^0) + g'^0(x - x^0)$ for all $x^0 \geq 0$ and $x \geq 0$. Hence, for any $w^0 \geq 0$,

$$\begin{aligned} \phi(x, w^0) &= g(x) + w^0 F(x) \leq g(x^0) + w^0 F(x^0) + \\ &(g'^0 + w^0 F'^0)(x - x^0) = \phi(x^0, w^0) + \phi'_x(x - x^0) \end{aligned}$$

That is, condition (3) holds for all $x^0 \geq 0$ and $x \geq 0$. Under these circumstances, theorem 1 and theorem 2 combine to make conditions (1) and (2) both necessary and sufficient that x^0 and w^0 provide a solution for the problem.

IV. MAIN RESULTS

Let $F(x) = c - Ax$, then the capacity constraint of problem P can be rewritten as $F(x) \geq 0$. Using the theorem in section 3, the problem P can be transformed to the following conditions.

(4) $v'U(x) = \min[U(x)]$ where v is a nonnegative vector with unit component sum.

For $\phi(x, w) = v'U(x) + w'F(x)$, we have

$$(5) \quad \phi_{x_s} = v_s U'_s(x_s) - \sum_{l \in L_s} w_l = 0 \text{ for } s = 1, 2, \dots, S$$

$$(6) \quad w_l \geq 0 \quad \sum_{s \in S_l} x_s \leq c_l \quad w_l (\sum_{s \in S_l} x_s - c_l) = 0 \text{ for}$$

$$l = 1, 2, \dots, L$$

For each source s , let $w^s = \sum_{l \in L_s} w_l$ is the path bandwidth price that s faces. From condition (5), we have

$$v_s U'_s(x_s) = \sum_{l \in L_s} w_l = w^s, \text{ so } U'_s(x_s) = w^s / v_s, \quad (7)$$

$x_s = [U_s'^{-1}(w^s / v_s)]_{m_s}^{M_s}$ is the stable point of source algorithm.

We using the following source rate control algorithm

$$x_s(t+1) = [x_s(t) + k_s(U'_s(x_s(t)) - (w^s / v_s))]^+$$

Here $k_s > 0$ is a step size, v is determined by condition

$$v'U(x) = \min[U(x)].$$

On the other hand, link prices are adjusted in opposite direction to the

gradient $\phi_w(x, w) = c_l - x^l(w)$, where

$$x^l(w) = \sum_{s \in S_l} x_s(w), \text{ that is the aggregate source rate at}$$

link l .

Then we have the following price adjustment algorithm for

$$\text{link } l \in L \quad w_l(t+1) = [w_l(t) - \gamma(c_l - x^l(w))]^+ \quad (8).$$

Here $\gamma > 0$ is a step size, and $[z]^+ = \max\{z, 0\}$.

Following [7], we can get a synchronous source-link algorithm.

Source s 's algorithm:

At times $t = 1, 2, \dots$, source s :

1. Receives from the network the sum $w^s(t) = \sum_{l \in L_s} w_l(t)$ of link prices in its path.
2. Choose a new transmission rate $x_s(t+1)$ for the next period:
$$x_s(t+1) = [x_s(t) + k_s(U'_s(x_s(t)) - (w^s / v_s))]^+$$
3. Communicates new rate $x_s(t+1)$ to links $l \in L_s$ in its path.

Link l 's algorithm:

At times $t = 1, 2, \dots$, link l :

1. Receives rate $x_s(t)$ from all sources $s \in S_l$ that go through link l .
2. Computes a new price
$$w_l(t+1) = [w_l(t) - \gamma(c_l - x^l(w))]^+ \text{ where}$$

$$x^l(w) = \sum_{s \in S_l} x_s(w)$$
3. Communicates new price $p_l(t+1)$ to all sources $s \in S_l$ that use link l .

Similarly, we can also design an asynchronous source-link algorithm. The detail we are not presented here.

Remarks:

1. The parameter v_i corresponding to source i is the product of path price and marginal utility. That is, it is not

only mathematical meaningful, but also physical meaningful in real networks.

2. When the utility of all users are equal our algorithm is equivalent to bandwidth max-min allocation. In this special case, the parameter v_i becomes the total link price for all links in the path of source i .

V. NUMERICAL EXAMPLE

Consider the classical linear network with 2 links having different link capacities $c_1 < c_2$, and 3 sources, indexed by $i = 1, 2, 3$, shown in the following figure. Source 1 goes through both of the two links and the other two sources go through only one link, with source 2 goes through link 1 and source 3 goes through link 2.

Let all sources have the same utility function, that is to say, let $U_1(x) = U_2(x) = U_3(x)$. Then the utility max-min fairness here becomes the traditional max-min fairness. The authors in [8] calculated the source rates under max-min fairness:

$$x_3 = c_2 - c_1/2, x_2 = x_1 = c_1/2$$

The network throughput under this fairness criterion is $c_2 + \frac{c_1}{2}$.

We will use our algorithm to get this allocation to verify the correctness of our method.

First, we write down the optimization form of this problem

$$\text{Max min } (x_1, x_2, x_3)$$

$$\text{Subject to } x_1 + x_2 \leq c_1, x_1 + x_3 \leq c_2$$

The equivalent conditions for this problem are as following:

There exists some nonnegative v satisfy condition $v_1 x_1 + v_2 x_2 + v_3 x_3 = \min(x_1, x_2, x_3)$

$$v_1 + v_2 + v_3 = 1. \quad (i)$$

There exists x and some nonnegative w such for $\phi(x, w) = v_1 x_1 + v_2 x_2 + v_3 x_3 - w_1(x_1 + x_2 - c_1) - w_2(x_1 + x_3 - c_2)$

$$\phi_{x_1}(x, w) = v_1 - w_1 - w_2 = 0$$

$$\phi_{x_2}(x, w) = v_2 - w_1 = 0 \quad \phi_{x_3}(x, w) = v_3 - w_2 = 0 \quad (ii)$$

$$\phi_{w_1}(x, w) = c_1 - x_1 - x_2 \geq 0 \quad \phi_{w_2}(x, w) = c_2 - x_1 - x_3 \geq 0$$

$$w_1(c_1 - x_1 - x_2) = 0 \quad w_2(c_2 - x_1 - x_3) = 0 \quad (iii)$$

$$w_1 \geq 0, w_2 \geq 0$$

There are four cases for w_1, w_2

Case (1) if $w_1 = w_2 = 0$, then from

(ii), $v_1 = v_2 = v_3 = 0$, this contradict the condition (i)

Case (2) if $w_1 = 0, w_2 \neq 0$, then from (ii) $v_1 = v_3 = w_2$,

$v_2 = 0$, with (i) we get $w_2 = 1/2, v_1 = v_3 = 1/2$, put these numbers into (ii), we get $\frac{1}{2}x_1 + \frac{1}{2}x_3 = \min(x_1, x_2, x_3)$. This condition holds if and only if $x_1 = x_3 \leq x_2$

From the second equation in (iii), we have $x_1 = x_3 = \frac{c_2}{2}$, the first capacity constrain requires $c_1 - x_1 - x_2 \geq 0$, that is, $x_2 \leq c_1 - \frac{c_2}{2}$. Combine these three conditions we get $c_2 \leq c_1$, which is contract to the problem.

Case (3) if $w_1 \neq 0, w_2 = 0$, similar as the above case, we get $v_1 = v_2 = w_1 = \frac{1}{2}, v_3 = 0, x_1 + x_2 = c_1$. Uses these numbers to replace the variables in condition (i), we get $\frac{1}{2}x_1 + \frac{1}{2}x_2 = \min(x_1, x_2, x_3)$. This condition holds if and only if $x_1 = x_2 \leq x_3$, so $x_1 = x_2 = \frac{c_1}{2}, x_3 = c_2 - \frac{c_1}{2}$. This satisfies all conditions, so this is the allocation achieved max-min fairness.

Case (4) if $w_1 \neq 0, w_2 \neq 0$, then from (iii) we have $x_1 + x_3 = c_2, x_1 + x_2 = c_1$. From these two conditions and $c_2 > c_1$, we know $x_3 > x_2$. From (ii) we get $v_1 = \frac{1}{2}, v_2 + v_3 = \frac{1}{2}$, $\frac{1}{2}x_1 + v_2 x_2 + v_3 x_3 = \min(x_1, x_2, x_3)$ because $v_1 \neq 0, v_2 \neq 0, v_3 \neq 0$, so the above equation never holds.

Even if the sources have different utility functions, the traditional max-min fairness allocation still results to this. Because this fairness criterion just takes into account the bandwidth allocated to each application, with no regard to the difference of users' utility functions. This criterion also takes no account of usage of network resources.

This disadvantage and some other implementation reasons of max-min fairness result in a new notion of fairness criterion introduced by Kelly [6]. Informally, rate allocations x_r are proportionally fair if they maximize $\sum_{r \in R} \log x_r$ under the capacity constraints. This objective may be interpreted as being to maximize the overall utility of rate allocations assuming each source has a logarithmic utility function. This fairness takes into consideration the usage of network resources and penalizes long routes using more resources more severely than max-min fairness in the interest of greater overall network throughput. For this simple example, it is not hard to solve directly to obtain the source rates under proportional fairness:

$$x_1 = \frac{1}{3}(2c_1 - c_2 + \sqrt{c_1^2 + c_2^2 - c_1 c_2})$$

$$x_2 = \frac{1}{3}(c_1 + c_2 - \sqrt{c_1^2 + c_2^2 - c_1 c_2})$$

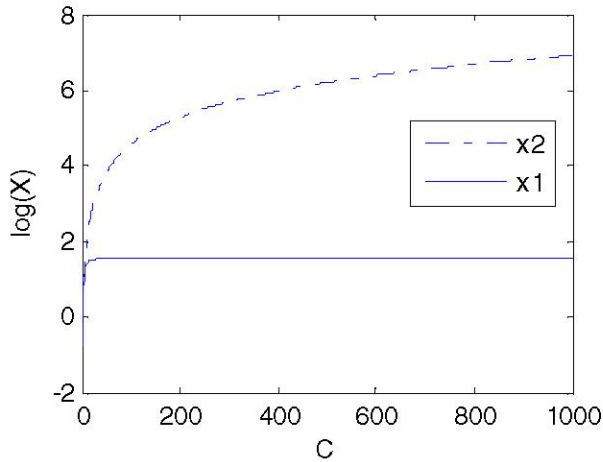
$$x_3 = \frac{1}{3}(2c_2 - c_1 + \sqrt{c_1^2 + c_2^2 - c_1 c_2})$$

Hence the network throughput under this fairness criterion is $\frac{2}{3}c_1 + \frac{2}{3}c_2 + \frac{1}{3}\sqrt{c_1^2 + c_2^2 - c_1c_2}$, which is greater than the throughput under the max-min fairness.

Now we use the following utility functions for these three sources and use our algorithm to derive their source allocations to compare with the above two situations.

Let source 1 has utility function $U_1(x_1) = \log(x_1)$, source 2 has utility function $U_2(x_2) = \arctan(x_2)$, source 3 has utility function $U_3(x_3) = \arctan(x_3)$. Using our method, we can get the source allocations for different link capacity.

The following figure shows the change of source rates as the link capacity varies.



The x-axis is the capacity of link 1. The y-axis is the logarithm of source rate.

As the link capacity increases, the rates of source 1 and source 2 both increase dramatically until some point is reached, then rate of source 1 almost remain constant, but rate of source 2 increases as before. This is because the marginal utility of source 1 is less important than source 2. That is, under same condition, the increasing rate of source 2 results in great throughput.

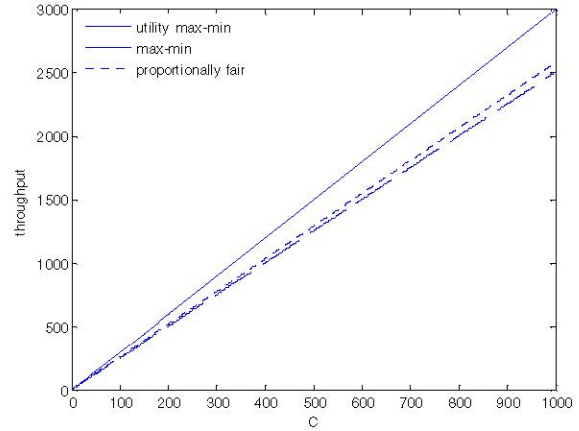
Under this fairness criterion, if we give proper utility function to flows which use more resources, then the network penalizes these flows using more resources more severely than proportional fairness.

Thus the network throughput under this fairness criterion would be greater than proportional fairness.

This fact is confirmed by our simple computation. The following figure confirms this conclusion.

VI. CONCLUSION AND FUTURE WORK

In this paper, we propose a new algorithm to achieve utility max-min fairness in bandwidth allocation problem. Future research will be carried out on simulations of this source-link algorithm and the proof that the algorithm indeed converges to the optimal solution. The stability of this scheme should be considered in future work.



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